

Data interpretation

Refer to the following table showing the production possibilities for a hypothetical economy.

Good	A	B	C	D	E	F
Consumption Goods	0	200	400	600	800	1000
Capital Goods	250	200	150	100	50	0

- What is the opportunity cost of increasing consumption goods production by 100 units? What is the opportunity cost of shifting production from point D to point C? (2 marks)
- Distinguish between a consumption good and a capital good - provide an example of each. (4 marks)
- Draw a Production Possibility Frontier diagram using the data from the table. Show how the PPF model can be used to illustrate economic growth. (4 marks)

Extended writing

Answer each of the following questions in about a page of normal writing. Use diagrams and examples where appropriate. Each question is worth 10 marks.

- 'The government should provide free health services to everyone in society.'
Evaluate this statement using the concepts of scarcity, opportunity cost, the production possibility frontier.
- Explain, using diagrams where relevant, how the production possibility frontier can be used to explain the following:
 - opportunity cost
 - the scarcity of resources
 - economic growth
 - changes in technology
- What is an economic model? Why do economists use theories and models in economic analysis?
- In most Australian cities, water restrictions have been imposed to deal with the shortage of water. Perth has built desalination plants to help increase the supply of water. Write an essay on the different options to overcome the scarcity of water. Are water restrictions fair? Should the price of water be increased? Would people use less water if the price was increased? Would this be fair? Should a pipeline be built from the Ord River, where water is plentiful, to Perth?

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Demand and supply

**Learning Objectives**

In this chapter you will learn about:

- different types of markets
- the law of demand
- the factors affecting demand
- the law of supply
- the factors affecting supply
- the concept of market equilibrium
- the price mechanism and how it clears market surpluses and shortages
- the effect of changes in demand and supply on market equilibrium

The role of markets

Economic systems exist to deal with the economic problem. An economic system tries to coordinate the production and distribution of goods and services to satisfy the wants of society. Economic systems vary from 'free market' to 'planned', but regardless of the type they must all answer the following fundamental questions:

- **What** goods and services will be produced and in what quantity?
- **How** will the goods and services be produced in terms of production technique and types of resources?
- **For whom** will the goods and services be produced?

In this chapter, we outline how these questions are answered in a market economy. Markets are one way to coordinate economic activity - they allocate resources in order to resolve the economic problem. Markets help to allocate scarce resources to satisfy the many competing wants in an economic system. Markets coordinate economic activity and provide incentives for both consumers and producers. Economists believe that markets are an efficient way to deal with the economic problem because they do not require any resources to function. The market coordinates decisions between buyers and sellers in a decentralised way. Markets work because when people engage in exchange, they gain!

A **market** is said to exist when buyers and sellers exchange goods, services or resources. A market consists of three important elements:

- buyers (demand)
- sellers (supply)
- something to exchange (a good, service or resource)

A market is also characterised by voluntary exchange. This means that buyers and sellers are not forced to participate in exchange, but do so willingly. This is important because buyers and sellers might meet face-to-face, but could also exchange goods and services by mail, telephone, or the internet. Buyers and sellers may also deal with one another through an intermediary or middle-man, as happens in the stock market, where transactions are usually handled by brokers on behalf of their clients.

Product markets deal with the buying and selling of goods and services. In a product market, consumers represent the demand side of the market, while producers or firms represent the supply side. Firms produce goods and services and sell them to households (consumers). **Factor markets** deal in the buying and selling of factors of production or resources such as the labour market, the capital market and natural resource markets. In a factor market, households sell their resources to firms. In a factor market, households represent the supply side of the market, while firms represent the demand side.

Markets can also be classified according to the intensity of competition in the market. A **competitive market** is characterised by:

- a large number of buyers and sellers

- firms are price takers
- very similar (homogeneous) products
- easy entry into the market (no barriers to entry or exit)

In a competitive market, price is determined by the interaction between buyers and sellers. No individual buyer or seller can influence the market price - this means that there is no market power. Firms are called price takers because they must take the price that is established by the market. There are many examples of competitive markets - agricultural markets, fruit and vegetables, cafes and hair dressing salons. The small business sector satisfies many of the assumptions of a competitive market. Any market with a large number of relatively small firms selling very similar goods and/or services and with easy entry conditions is a competitive market. A market that is not competitive is called an **imperfect** or non-competitive market. An imperfect market is characterised by:

- a small number of firms
- product differentiation
- firms are price setters - they have market power
- entry into the market is restricted

A monopoly market is an extreme type of imperfect market with just one dominant firm. An oligopoly is a market with a few dominant firms. Firms with market power can set their own price - they are called price setters. An imperfect market is characterised by product differentiation where firms try to make their products 'look different' in the eyes of consumers and use barriers to entry to make it difficult for new firms to enter the market. In Australia there are many examples of imperfect markets. Coles and Woolworths are a virtual duopoly in the grocery market. The mobile phone services market is a classic oligopoly dominated by three firms: Telstra, Optus and Vodafone.

The role of prices

In the market system the three key economic questions are answered by the price mechanism - the interaction of buyers and sellers in the market. In a market economy, the consumer is king (consumer sovereignty), meaning that consumers determine what will be produced, and how much will be produced. Consumers signal their tastes and preferences to producers by casting their dollar votes when they buy goods and services. Producers respond to changes in demand signalled by consumers. For example, if striped shirts become more popular than plain shirts, then shirt producers would increase the production of striped shirts and reduce the output of plain shirts. In the motor vehicle market sales of SUVs have soared while sales of large sedans have slumped as a result of changes on consumer tastes.

In the market economy, the how to produce question is answered by producers. Like consumers, producers are guided by the principle of self interest. To operate profitably, firms have to operate efficiently and minimise costs. Producers will compare the prices of inputs when deciding what techniques of production to use.

The market system provides an incentive for the producer to become more efficient and lower costs in order to increase profits. Price also determines the 'for whom' question. The rewards of production reflect the price which the market is willing to pay for resources. Wages and salaries are the reward for labour; owners of natural resources receive rent; the owners of capital receive dividends (or interest in the case of money capital); and the owners of entrepreneurial skills receive profits. The amount of income earned by resource owners generally reflects the scarcity of the resource they have provided. Skilled workers, for example, are paid a higher wage than unskilled workers. Doctors and engineers will receive a higher income than clerical staff or sales staff. Musicians and entertainers such as Taylor Swift and Ed Sheeran earn millions of dollars because of their special skills. We will now turn our attention to how competitive markets determine prices. We will focus on the most important model in microeconomics - the model of demand and supply.

The demand side of the market

Demand refers to the buying intentions of consumers. The quantity demanded of a good or service is the quantity that consumers are willing and able to purchase, at a particular price and at a particular time. Demand is not the same thing as a want. Many people may want a luxury yacht, but only a few are willing and able to purchase the yacht. Demand refers to the actual buying intentions of consumers. I might want a Ferrari, but I actually purchase a Mazda 3 because that is what I can afford.

An important law in economics is the law of demand - as the price of a good rises, people buy less of it, *ceteris paribus* (holding other factors constant).. The law of demand describes a negative relationship between price and quantity demanded. What is the explanation for the law of demand? There are two important reasons.

- The **income effect** - when the price of a good rises, consumers are not willing to buy as much of the good because their real income or purchasing power has decreased. If you have \$100 in income and the price of a pizza is \$10 then your real income is 10 pizzas. If the price of pizzas increases to \$20, then your real income halves to just 5 pizzas. When your real income falls you are poorer and therefore you will purchase less. The income effect works in both directions - when the price of a good falls, your real income rises and you will purchase more.
- The **substitution effect** - when the price of one good rises, other goods become more attractive to buyers because they are relatively cheaper. An increase in price will cause consumers to switch to relatively cheaper substitutes. For example, an increase in the price of pizza will cause some consumers to switch to other types of fast food. On the other hand, a decrease in the price of pizza, will now make substitute goods relatively more expensive and consumers will purchase more pizza.

Notice that both the substitution effect and the income effect work together to reinforce the law of demand. Are there any exceptions to the law of demand? The

simple answer is No! The law of demand holds for all goods and services - as long as the **ceteris paribus** (keeping other factors constant) condition holds. Don't be fooled by people who claim that there are exceptions to the law of demand. For example, it is often argued that if a consumer buys an expensive brand rather than a cheap brand, this violates the law of demand. This is incorrect because we are dealing with two different goods. The consumer may use price as a signal of quality and have a preference for the more expensive brand. To break the law of demand, the consumer when given a choice of buying the same good at two different prices chooses to pay the higher price. When buying a good, would you refuse a discount? Of course not! There are no exceptions to the law of demand.

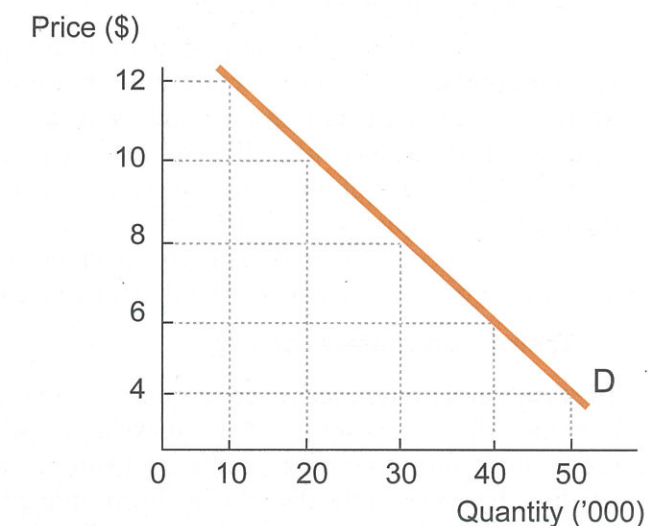
Of course, the law of demand varies in strength between different types of goods depending on how strong the income and substitution effects are. For example, the law of demand for petrol will be relatively weak. There are few close substitutes for petrol. If the price of petrol increases, consumers will find it difficult to substitute away from petrol. The law of demand for orange juice, on the other hand, will be much stronger. There are many close substitutes for orange juice such as apple, pineapple and grape juice. If the price of orange juice increases, then consumers can readily substitute into another fruit juice.

Figure 2.1 shows a demand schedule and a demand curve for pizza. A demand schedule is simply a table showing the quantity demanded of a good consumers are willing to purchase at various prices. Figure 2.1 shows the weekly quantity of pizzas that consumers are willing to buy at various prices. It is called a market demand schedule because it is the summation of all the individual consumers in the market. As we would expect, rational consumers are willing to buy more pizzas at lower prices than they are at higher prices.

Figure 2.1 The demand curve

The demand schedule and graph illustrate the Law of Demand - price and quantity demanded are negatively related, *ceteris paribus*. As price rises, quantity demanded falls; as price falls, quantity demanded rises.

Demand schedule	
Price (\$)	Quantity demanded
12	10000
10	20000
8	30000
6	40000
4	50000



We can convert the data from the schedule to a graph known as the market demand curve for pizza. The graph's axes represent price on the Y axis, and quantity demanded on the X axis. Note that the demand curve is negatively sloped (downward sloping to the right) reflecting the law of demand. As price falls, quantity demand rises and when price rises, quantity demanded falls. There are other factors apart from price, which may affect the demand for a good or service. These are all the factors that were held constant (*ceteris paribus* conditions) in deriving the demand curve. These non-price factors affecting demand are discussed in the next section.

Non-price factors affecting demand

What are the key factors, other than price, that can affect demand?

- **Levels of disposable income**

Your level of income determines your budget - what you can and cannot afford. Consumers will normally purchase more of a good when their income increases. Most goods in the economy are referred to as normal goods. A normal good is defined as one where demand increases as income increases. An inferior good is a good where demand decreases as income rises. An example of an inferior good is a very low quality product such as a 'plain wrap' or 'home' brand product.

- **The price of related goods**

On most occasions, consumers can choose between a number of goods and services which satisfy the same wants. These are referred to as **substitutes**. For example, there is a wide range of fast food products on the market other than pizza. Consumers can choose from chicken, hamburgers, kebabs, noodles, or fish and chips - the list is very long and varied. If the price of pizza rises, the quantity demanded of pizza will fall and consumers will increase their demand for other types of fast food. There are many other examples of substitute goods such as tea and coffee, butter and margarine, orange and apple juice.

Many goods are also consumed with other goods - these goods are called **complements**. For example, computers require software; filing cabinets require document folders; cars require petrol; and the playing of tennis requires courts, racquets and balls. You might like to consume pizza with garlic bread or with a bottle of coke. If the price of pizza rises, the quantity demanded of pizza will fall, and the demand for complement goods, such as garlic bread will decrease. Similarly, a rise in the price of computers will lead to a decrease in the demand for software while a decrease in the price of cars will lead to an increase in the demand for petrol.

- **Tastes and preferences**

The tastes and preferences of consumers are an important determinant of consumer demand. The advertising industry plays an important role in affecting consumers' preferences and spending habits. Consumers have quickly embraced the new products/services of the digital age - from smartphones to Facebook and Uber. Sales of CDs quickly declined as consumer preferences shifted towards digital recordings.

Similarly, demand for DVDs has declined as consumers switch to online streaming of movies. Many consumers now indicate a preference for environmentally friendly 'green' and organic products.

- **Expectations of consumers**

If people expect conditions to change in the future, they may make decisions now rather than postpone them. For example, if the price of petrol is expected to rise in the future, then consumers will have an incentive to increase their purchases now to beat the price rise. This is rational behaviour. Expectations of a future price rise will increase demand now and shift the demand curve to the right. If you expected the price of petrol were to fall next week, then you would decrease your current demand and wait for the price to fall.

- **Demographic factors**

The size and age composition of the population can have an important bearing on the pattern of demand. A growing population increases the market size for all goods and services whereas a change in the age profile will affect specific goods and services. For example, as the Australian population ages, the demand for child care facilities will decrease relative to the demand for retirement facilities. There is a growing demand for ocean cruises as the population ages and more people enter retirement age.

Changes in demand

We distinguish between two types of changes in demand:

- a movement along the curve - caused by a change in price.
- a shift of the entire curve - caused by a non-price factor.

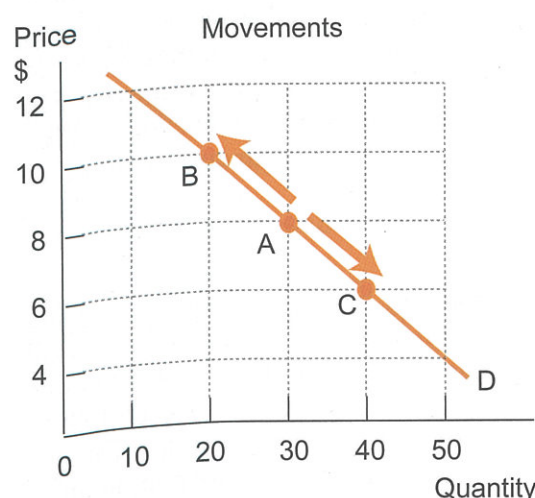
These changes are shown in figure 2.2. A change in the price of the good itself will lead to a movement along the demand curve. A change in price will cause a change in quantity demanded. For example, an increase in the price of a good will cause a decrease in quantity demanded whereas a fall in price will cause an increase in quantity demanded (panel A). It is important to remember that when price changes, we move along the curve. A change in price cannot shift the curve.

Changes in any of the other non-price (*ceteris paribus*) factors affecting demand will cause a shift in the entire demand curve. When non-price factors change, the result is a new demand curve. There are two types of shifts - an increase in demand occurs when the demand curve shifts to the right; a decrease in demand occurs when the demand curve shifts to the left (refer to panel B in figure 2.2). An increase in demand, for example, might result from any of the following events:

- an increase in consumer income for a normal good
- a decrease in consumer income for an inferior good
- a fall in the price of a complementary good

Figure 2.2 Changes in demand

A. A change in price causes a movement along the curve. A rise in price decreases qty demanded, while a fall in price increases qty demanded.



B. Increases or decreases in demand are shifts of the entire curve, caused by a factor other than the price of the good.



- An increase in the price of a substitute good
- A change in tastes and preference in favour of the good
- The expectation that price will rise in the future
- An increase in the number of consumers in the market

The supply side of the market

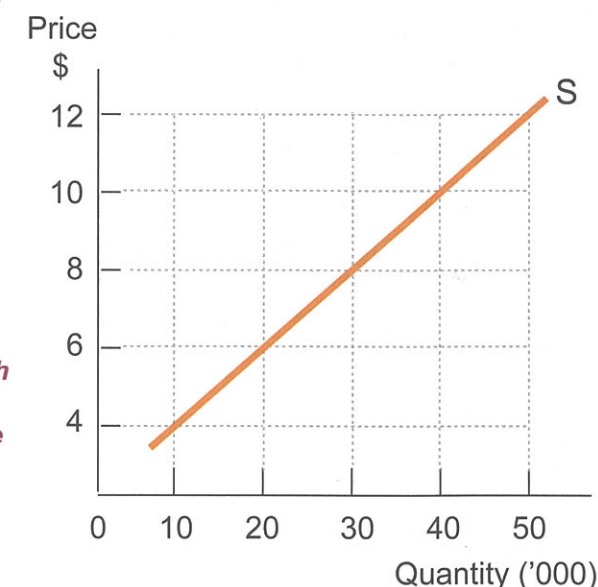
Supply represents the sellers or producers side of the market. Supply is the amount of a good or service that producers are willing and able to sell at a particular price and at a particular point in time. The relationship between the quantity supplied and the price of a good is a positive one. The law of supply states that as the price of a good or service rises, the quantity supplied will also rise. Notice that this is opposite to the law of demand. The law of supply also assumes that all non-price factors that may affect supply are held constant - the *ceteris paribus* assumption. What is the economic reason for the law of supply? Rational, self-interested producers would prefer to sell their output at a high price than at a low price because of the profit motive. If you are selling a good or service you would prefer to sell at the highest price possible. When firms increase their output they may also find that their production costs may increase which would require price to rise to maintain their profit margin.

Figure 2.3 depicts the supply schedule and the supply curve for pizzas. A supply schedule is simply a table showing the quantity supplied of a good producers are willing to sell at various prices. Figure 2.3 shows the weekly quantity of pizzas that

Figure 2.3 The supply curve

Supply schedule	
Price (\$)	Quantity supplied
12	50000
10	40000
8	30000
6	20000
4	10000

The supply schedule and graph illustrate the Law of Supply - price and quantity supplied are positively related, other things being equal. As price rises, quantity supplied rises.



producers are willing to sell at various prices. It is called a market supply schedule because it is the summation of all the individual producers in the market. As we would expect, rational producers are willing to sell more pizzas at higher prices than they are at lower prices. We can convert the data from the schedule to a graph known as the market supply curve for pizza. Note that the supply curve is positively sloped, reflecting the law of supply.

Non-price factors affecting supply

There are other non-price factors which can affect the supply for a good or service. These are all the factors that were held constant (*ceteris paribus* conditions) in deriving the supply curve. The most important factors are the costs of production; technology; prices of other goods the firm can produce; the number of sellers; and expectations of producers.

• Costs of production

The price of resources such as labour, capital and raw materials determine the firm's costs of production. A supply curve can be thought of as a cost curve for the firm. Usually, a firm's largest costs are wages and salaries and the cost of purchasing raw materials. An increase in production costs will mean that for a given price, the firm will decrease its production. This will cause the supply curve to decrease - to shift to the left.

• Technology

Technology is knowledge about the techniques of production. If technology improves, then more output can be produced from the same quantity of resources. This means that the firm's costs of production will fall. Other things being equal, an

improvement in technology will result in an increase in supply. The supply curve will shift to the right. The cost of many products including computers, household appliances and electronic devices has continuously fallen over time as a result of improvements in technology.

- **Prices of other goods**

Production is a process which involves combining productive resources to produce a final good. In many cases the same resources could be used to produce other goods. The producer will closely monitor movements in the prices of the goods he is capable of supplying so that he can take advantage of profit opportunities. In agriculture, for example a farmer may shift resources from wheat production to wool if the price of wool rises relative to wheat. A car manufacturer will decrease the supply of large cars if small cars become more popular.

- **Number of sellers**

If new sellers enter the market, then market supply will increase and the market supply curve will shift to the right. Sellers might be enticed to enter a market if profit opportunities are strong.

- **Expectations of producers**

Suppliers expectations of future conditions will affect supply. If a higher price is expected in the future, then firms will decrease their current supply in order to take advantage of future higher prices. Firms will make decisions about investing in plant and machinery on expected future prices. Expected high prices for oil for example, will encourage increased exploration.

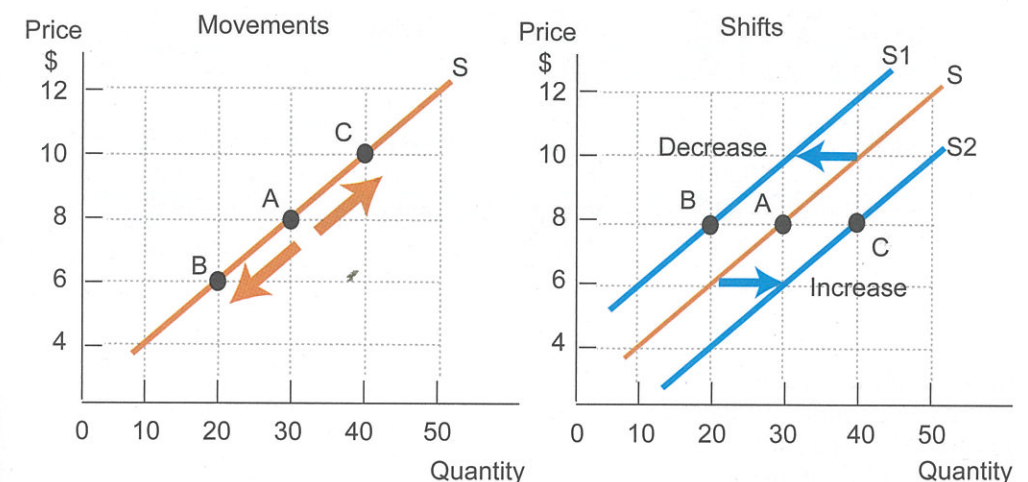
Changes in supply

There are two types of changes in supply:

- A movement along the supply curve - caused by a change in price
- A shift of the entire supply curve - caused by a change in a non-price factor

These changes are shown in figure 2.4. Changes in the price of the good itself will lead to a movement along the supply curve. A change in price causes a change in quantity supplied. An increase in the price of the good will cause an increase in the quantity supplied while a fall in price leads to a decrease in the quantity supplied. It is important to remember that when price changes, we move along the curve. A change in price cannot shift the curve. Changes in any of the other non-price factors affecting supply will cause a shift in the whole supply curve. These factors were all held constant when the initial supply curve was drawn. Allowing these factors to change will result in a new supply curve. There are two types of shift - an increase in supply occurs when the supply curve shifts to the right; a decrease in supply occurs when the supply curve shifts to the left. An increase in the supply of a good will result from any of the following events:

Figure 2.4 Changes in supply



A change in price causes a movement along the curve. A rise in price increases quantity supplied, while a fall in price decreases quantity supplied.

Increases or decreases in supply are shifts of the entire curve, caused by a factor other than the price of the good.

- an improvement in technology
- a decrease in production costs (input prices)
- the expectation that prices will fall in the future
- an increase in the number of producers in the market

Equilibrium price and quantity

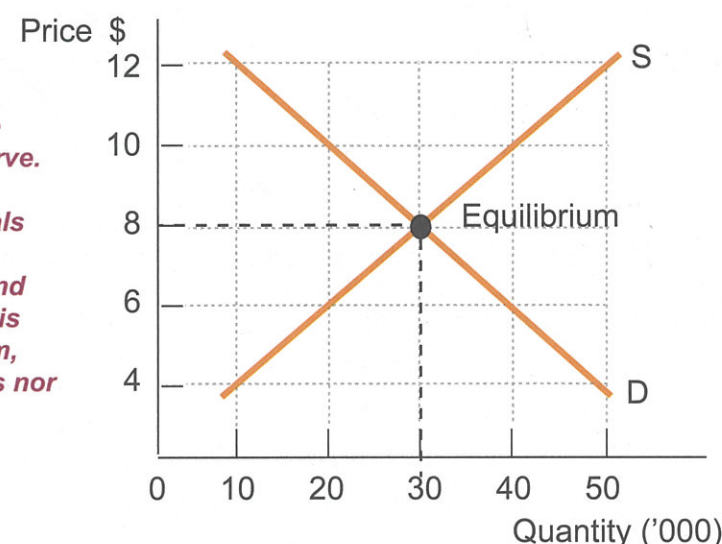
It is important to remember that a market consists of both buyers (demand) and sellers (supply). To determine the market price and quantity we combine the demand and the supply curve on the one graph. The price that clears the market is referred to as the **equilibrium price**. This is the price that balances the buying intentions of consumers with the selling intentions of producers. Figure 2.5 brings the two sides of the market (demand and supply) together. The equilibrium price and equilibrium quantity will be where quantity demanded equals quantity supplied. This will be the price where the demand curve intersects with the supply curve. At the equilibrium price, the amount that buyers are willing to buy is exactly equal to the amount that sellers are willing to sell. In our example of the pizza market, when the price for a pizza is \$8, buyers are willing to buy 30,000 and sellers are willing to sell 30,000. At any other price than \$8, there would be an imbalance between quantity demanded and quantity supplied.

What will happen if the market price is not at the equilibrium price? Will the market automatically adjust to the equilibrium price and quantity? Figure 2.6 shows the situation when the price is either above (panel A) or below (panel B) the equilibrium

Figure 2.5 Market equilibrium

Price \$	Quantity demanded	Quantity supplied	Result	Effect on price
12	10000	50000	surplus	fall
10	20000	40000	surplus	fall
8	30000	30000	equilibrium	no change
6	40000	20000	shortage	increase
4	50000	10000	shortage	increase

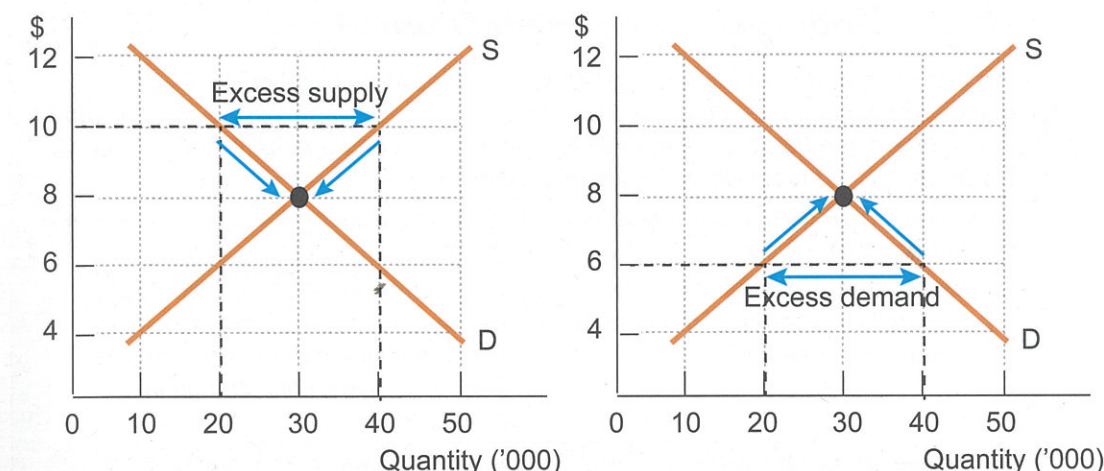
The equilibrium occurs where the demand curve intersects the supply curve. At the equilibrium price, quantity demanded equals quantity supplied. The equilibrium price is \$8 and the equilibrium quantity is 30,000. At the equilibrium, there is neither a surplus nor a shortage.



price. Suppose that the price of pizzas happened to be \$10. At this price, sellers are offering a quantity of 40,000 for sale, while buyers only wish to purchase 20,000. There is an excess supply or **surplus** of 20,000 pizzas. The price is too high to clear the market. Sellers will want to eliminate the surplus by lowering prices. As prices fall, more consumers will want to purchase so that the excess supply will be eliminated. Notice that as price falls, consumers move down along the demand curve, while sellers move down along the supply curve. As long as there is a surplus, sellers will have an incentive to lower prices - the price will keep falling until it reaches the equilibrium price of \$8.

Suppose that the price of pizzas happened to be \$6. At this price, sellers are offering a quantity of just 20,000 for sale, while buyers wish to purchase 40,000. Now there is an excess demand or **shortage** of 20,000 pizzas. The price is too low to clear the market. Buyers will react to the shortage by bidding up the price. As the price rises, more sellers will want to supply pizza and the excess demand will be eliminated. Notice that as the price rises, consumers move up along the demand curve, while sellers move up along the supply curve. As long as there is a shortage, sellers will have an incentive to increase prices - the price will keep rising until it reaches the equilibrium of \$8.

Figure 2.6 Surpluses and shortages



Only when the price of pizzas is \$8 will quantity demanded exactly match quantity supplied - 30,000 pizzas. At this price there is neither a surplus nor a shortage. The market is said to be in equilibrium. The word equilibrium means a state of balance or no tendency to change. Any other price/quantity combination will automatically cause both the price and quantity to change in the direction of equilibrium. Excess supply forces prices down while excess demand causes prices to rise. These pressures for change are known in economics as the invisible hand of the market - a term coined by Adam Smith. A more modern term is the price mechanism.

Prices are always adjusting in competitive markets in response to changes in demand and supply conditions. Changes to the market will create either temporary shortages or surpluses which will then cause price to change to re-establish a new equilibrium. The length of time required to eliminate shortages and surpluses though will vary from one market to another. In the share market, prices change very quickly to find a balance between buyers and sellers. The price of the Australian dollar in the foreign exchange market changes by the minute to balance supply and demand factors. In fruit and vegetable markets, prices adjust daily in response to supply and demand conditions. These are all good examples of competitive markets. As long as a market is able to operate freely, prices will always adjust to balance demand and supply.

Changes in demand and supply

Why do prices in markets fluctuate? The reason is that either demand conditions change which shifts the demand curve or supply conditions change which shifts the supply curve. Remember, a given demand curve and supply curve will establish a unique equilibrium price and quantity, but if the demand curve or the supply curve shift, then a new equilibrium will be established. It is important to understand what happens in the market given a change in either demand or supply. The explanatory box on page 38 highlights the four stage explanation process to understanding

Understanding changes in equilibrium

We can get a better understanding of how changes in demand or supply affect market outcomes (price and quantity sold) using a 'four-stage' explanation method. This method can be used to analyse any given change to a market and predict what will happen to both equilibrium price and quantity.

Example 1: How does the market for apples react to a report that says "an apple a day keeps the doctor away"

1.	Identify the cause of the change in demand or supply	Demand for apples increases (the D curve shifts to the right) because there is a favourable change in preferences
2.	Explain how the change causes a temporary shortage or surplus at the initial equilibrium	Increased demand creates a temporary shortage at the initial price
3.	Describe how market price reacts to the temporary shortage or surplus	Price always rises when there is a market shortage
4.	Describe how this causes a change in the quantity demanded or supplied	The increase in price causes an increase in Q_s and a decrease in Q_d as sellers and buyers react to the price rise

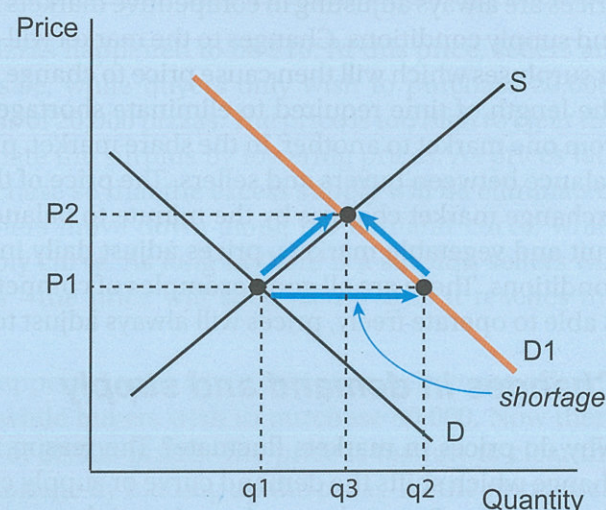
The apple market is in equilibrium at P_1 and q_1 . An increase in demand shifts the D curve to D_2 . At P_1 , Q_d increases to q_2 - but this creates a shortage = q_1q_2 . As price is bid up to P_2 , Q_s rises and Q_d falls to the new equilibrium of q_3 .

If there was a decrease in demand in a market, then there would initially be a temporary surplus - this would cause the price to start falling clear the surplus.

Use the 4 stage explanation to analyse the following:

1. The market for bananas is adversely affected by a cyclone.
2. A news report warns of the negative health effects of consuming coffee.

The Apple Market



changes in markets. We can refer to four types of changes to a market equilibrium, because both the demand curve and the supply curve can increase (shift to the right) or decrease (shift to the left):

- an increase in demand.
- a decrease in demand.
- an increase in supply.
- a decrease in supply.

For each of these changes, we need to analyse the impact on equilibrium price and quantity.

• Increase in demand

Imagine a new medical report that announced that pizza consumption improved your intelligence. This would lead to an increase in the demand for pizzas - the demand curve for pizzas will shift to the right, as consumers' preferences change (see Figure 2.7, panel A). At the original equilibrium price of \$8, quantity demanded will now exceed quantity supplied. This excess demand for pizzas will cause the price to rise to a higher equilibrium where the new demand curve intersects with the supply curve. At the higher price, quantity supplied will increase. So an increase in demand will result in an increase in the equilibrium price and equilibrium quantity. Whenever you see rising prices and rising sales in a market, then you know that demand for the good is increasing.

• Decrease in demand

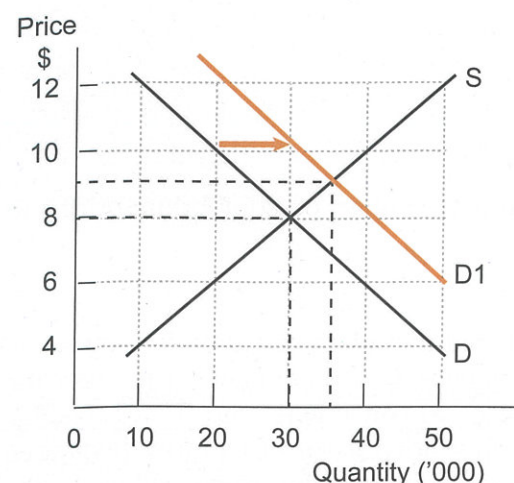
Imagine if it was reported that pizza consumption was a cause of cancer. Now consumers would change their preferences away from pizza. This would lead to a decrease in the demand for pizza - the demand curve for pizzas will shift to the left (see Figure 2.7, panel B). At the original equilibrium price of \$8, quantity supplied will now exceed quantity demanded. This excess supply of pizzas will cause the price to fall to a lower equilibrium. At the lower price, quantity supplied will fall. Whenever you see prices and quantities falling in a market, then you can safely assume that demand for the product is decreasing.

• Increase in supply

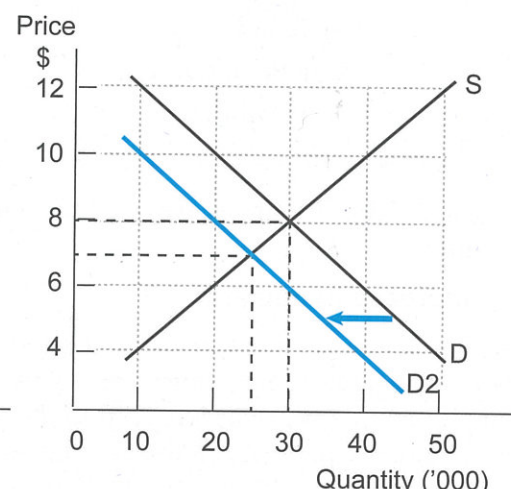
A change in production costs or technology are important factors that can shift the supply curve. Suppose that a new pizza oven is created which reduces the time to cook pizza helping to lower production cost. This will cause the supply curve to shift to the right - an increase in supply (see panel C). Pizza producers will now be able to supply more pizzas for the same cost. The equilibrium price will fall, while the equilibrium quantity will rise.

• Decrease in supply

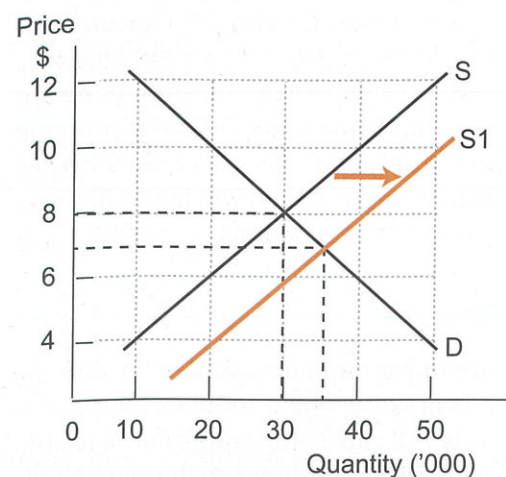
Suppose there was an increase in the price of mozzarella cheese - a key ingredient in producing pizzas. This would increase the cost of making pizza and would cause the supply curve to shift to the left - a decrease in supply (see panel D). The new

Figure 2.7 Changes in equilibrium**A. Increase in Demand**

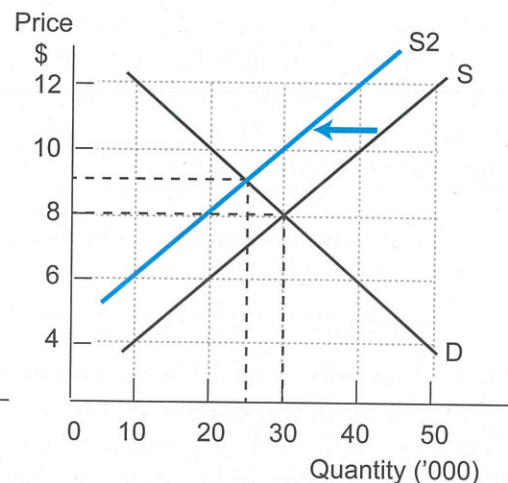
A. An increase in demand causes both equilibrium price and quantity sold to rise. Price has increased to \$9 and quantity has risen to 35,000 pizzas.

B. Decrease in Demand

B. A decrease in demand causes both equilibrium price and quantity sold to fall. Price has decreased to \$7 and quantity has fallen to 25,000 pizzas.

C. Increase in Supply

C. An increase in supply will cause a decrease in equilibrium price and an increase in equilibrium quantity - in this example, price falls to \$7, while quantity rises to 35,000.

D. Decrease in Supply

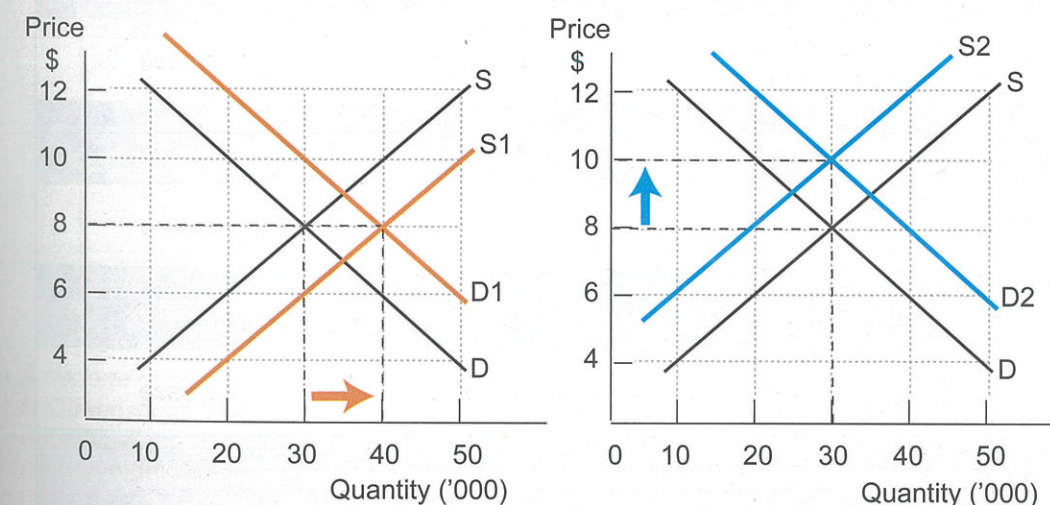
D. A decrease in supply will result in an increase in equilibrium price and a decrease in equilibrium quantity - in this example, price rises to \$9, while quantity falls to 25,000.

equilibrium will be where the new supply curve intersects with the original demand curve - price will rise while quantity sold will fall.

Simultaneous shifts in demand and supply

When a single curve shifts, our model of demand and supply makes it relatively easy to predict the movement in price and quantity. However, what will happen to equilibrium price and quantity if both curves shift at the same time? While this is more complex, our model can still be used to analyse the changes to price and quantity. For example, if we combine an increase in demand with an increase in supply (Figure 2.8, panel A), then the equilibrium quantity must increase, but the equilibrium price may rise, fall or stay constant - it is indeterminate. This is because an increase in demand pushes price up while an increase in supply pulls price down. What happens to price will depend on the relative shift of each curve. If the demand curve shifts more than the supply curve, then the equilibrium price will rise. If the supply curve shifts by more, then the equilibrium price will fall. In our example, both curves have shifted the same amount so that the equilibrium price has not changed.

Let's examine another case of a simultaneous shift. Panel B in Figure 2.8 shows the impact of an increase in demand combined with a decrease in supply. In this case equilibrium price must increase, but equilibrium quantity is now indeterminate. This is because an increase in demand increases quantity while a decrease in supply decreases quantity. Quantity may increase, decrease or remain constant - it will depend on the relative shifts of the curves. If the demand curve shifts more than

Figure 2.8 Simultaneous changes

A. An increase in demand and supply will cause quantity sold to rise but the effect on price is indeterminate - in this example, Q has increased to 40,000 while P has remained at \$8.

B. An increase in demand and a decrease in supply will result in price rising but the effect on quantity is indeterminate - in this example, P has increased to \$10 while Q has remained at 30,000.

the supply curve, then the equilibrium quantity will rise. If the supply curve shifts by more, then the equilibrium quantity will fall. In our example, both curves have shifted the same amount so that the equilibrium quantity has not changed.

Figure 2.9 summarises the effects on price and quantity of shifts in both demand and supply. Notice that when both curves shift, it is possible to only predict what will happen to either price or quantity but not both. Only if we have more information about the size of the relative shifts of each curve can we predict the outcome on both price and quantity. For example, if there was an increase in demand for pizza which was greater than a decrease in supply, then both price and quantity would increase.

Demand & Supply - Case Study

The Australian fruit and vegetable market is often affected by the cyclone season in Queensland - destructive winds and rain can destroy much of Queensland's crops resulting in shortages and high prices. The last cyclone to devastate Queensland destroyed a large part of the state's banana crop. Queensland accounts for around 95 per cent of Australia's banana production. What would we predict would happen to banana prices and sales? Using our model of demand and supply we would show the effect of the cyclone as a large decrease in supply. This would result in a large increase in banana prices and a significant decrease in banana sales (Figure 2.10).

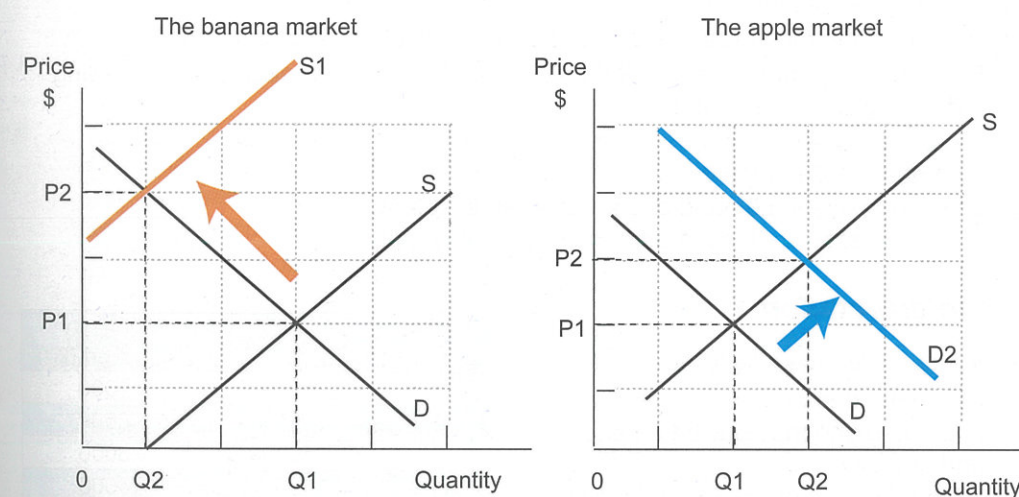
Figure 2.9 Predicting market changes

Type of Shift	Effect on Price	Effect on Quantity
Increase in demand and no change in supply	Increase	Increase
Decrease in demand and no change in supply	Decrease	Decrease
Increase in supply and no change in demand	Decrease	Increase
Decrease in supply and no change in demand	Increase	Decrease
Increase in demand and Increase in supply	Indeterminate	Increase
Increase in demand and Decrease in supply	Increase	Indeterminate
Decrease in Demand and Increase in Supply	Decrease	Indeterminate
Decrease in Demand and Decrease in Supply	Indeterminate	Decrease

Is there a shift in the demand curve? The answer is no! The supply curve shifts and we move up along the demand curve. If a news report told you that a cyclone was approaching, would you immediately have a craving for bananas? Would eating bananas protect you from a cyclone? The answer is no - the demand curve for bananas does not shift! Do not make the mistake of thinking that if banana prices increase this will cause a decrease in the demand for bananas and shift the demand curve to the left. It is important to remember that a change in price cannot shift the curve - only a change in one of the ceteris paribus factors can shift the curve.

What did banana consumers do in response to banana prices tripling? They obviously stopped buying bananas - as the price of bananas rose, quantity demanded fell (a movement up along the demand curve) and looked for a substitute fruit. Is there a close substitute for bananas? Yes, Australian consumers actually increased their purchases of apples! Apples certainly don't look or taste like bananas - they are generally red, round and crisp rather than yellow, long and squishy. But Australian consumers did switch to apples as their replacement fruit. In the apple market, there was a large increase in the demand for apples resulting in apple prices and sales both increasing. This is exactly what our model would predict.

Figure 2.10 The banana and apple markets



A cyclone destroys banana plantations in Queensland. This causes a large decrease in supply - increasing prices and decreasing quantity sold.

The rise in the price of bananas causes the demand for apples (a substitute good) to increase - apple prices and sales both increase.